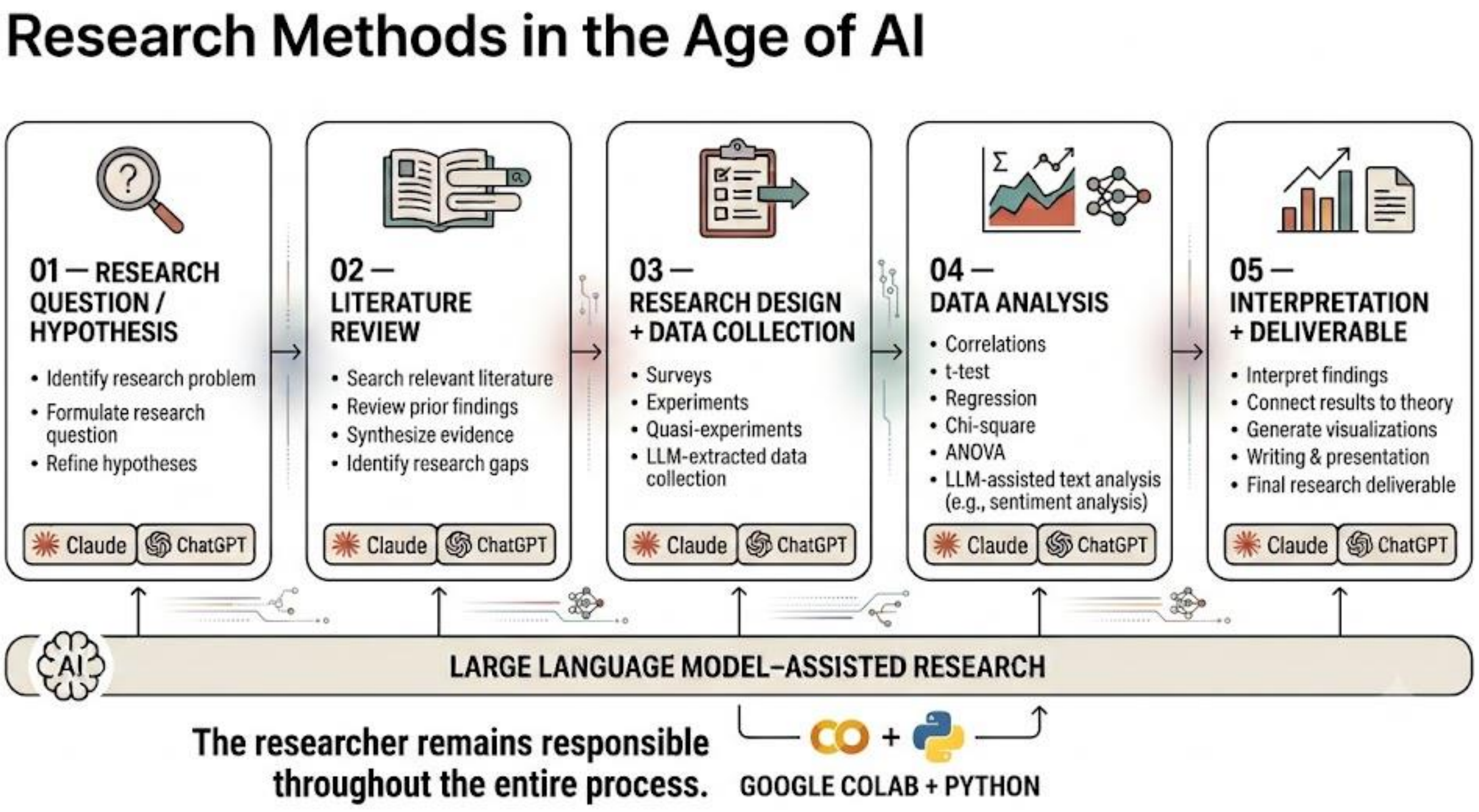
Python Fundamentals: Organizing and Processing Information

Why Learn Python?



Last week, we learned:

- What Python and Google Colab are
- Basic Python symbols
- Variables
- Functions
- Libraries

Recap: Reading Python Code

Symbol	Think of it as...	Simple example
`()`	**Do something / give something information**	`print("Hello")`
`[]`	**A group of things OR pick something from a group**	`names[0]`
`.`	**Use something that belongs to something else**	`text.upper()`
`" "` or `' '`	**This is text**	`"Hello"`
`,`	**Separate things**	`print("Kay", 10)`
`#`	**A note for humans — Python ignores it**	`# My analysis`
`=`	**Give something a value**	`age = 25`
`==`	**Ask: Are these equal?**	`age == 25`

Today, we will learn how Python can:

1. **Store** multiple pieces of information
2. **Find** information we need
3. **Repeat** a task automatically
4. **Make a simple decision**
5. **Save instructions to use again**

1. Python can store multiple pieces of information

“Lists” Store Information

- Lists can contain strings, numbers, mixed types, or even other collections.

```
headlines = [
    "AI changes journalism",
    "Election coverage expands",
    "New platform launches"
]
```

We Can Add or Change Items in a List

- A list doesn't have to stay the same.

```
#Change list (string)
headlines.append("Media trust declines")
```

```
#Change list (integer)
numbers = [1, 2, 3]
numbers[0] = 10

print(numbers)
```

“Dictionaries” also store information

- Think of a dictionary like one row of information
- In Excel:

Team	Sport	City
Gamecocks	Football	Columbia

- In Python:

```python
team = {
    "name": "Gamecocks",
    "sport": "Football",
    "city": "Columbia"
}
```

Dictionaries Store Labeled Information

- Key (i.e., label): Value (i.e., information)
- Written with { }
- Dictionaries are IMPORTANT for this course because they create a natural bridge to your understanding of data analysis AND LLM output later.

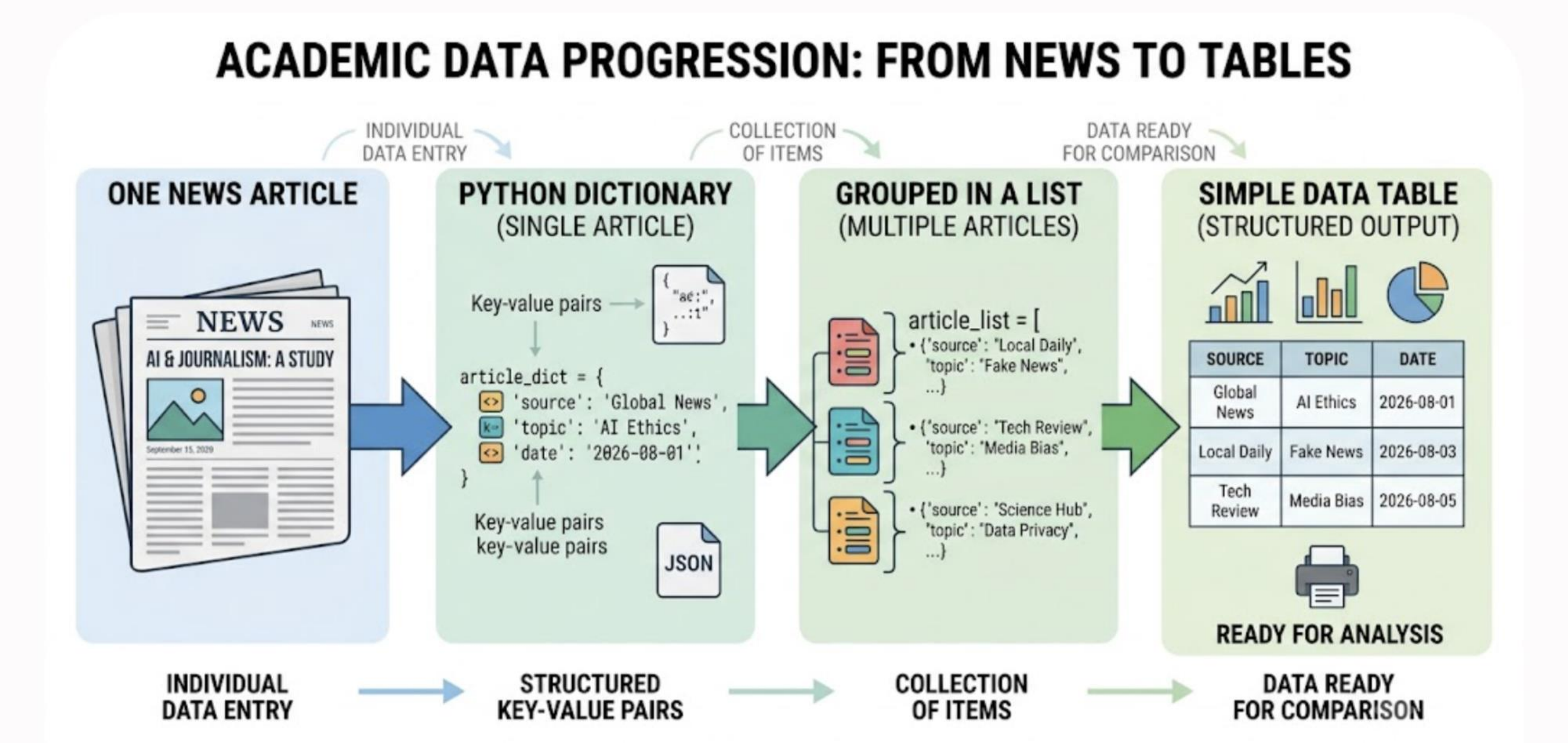
```
#Create a dictionary
article = {
    "source": "Daily News",
    "year": 2026,
    "topic": "AI",
    "sentiment": "positive"
}
```

We Can Put Many Dictionaries in a List

- One dictionary = information about ONE thing
- A list of dictionaries = information about MANY things

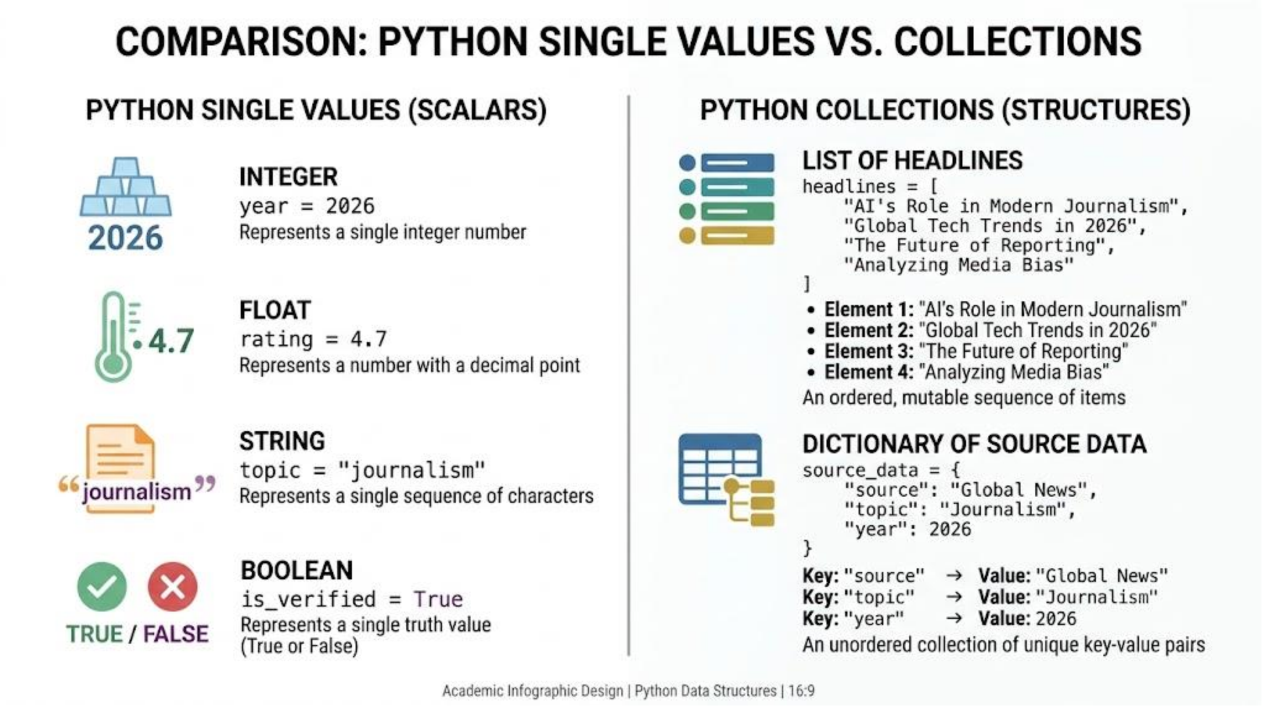
```
articles = [
    {"source": "CNN", "topic": "AI"},
    {"source": "NPR", "topic": "Politics"},
    {"source": "AP", "topic": "Health"}
]
articles
```

List of dictionaries → DataFrame



One Value vs. Many Values

- Sometimes we store **one value**
- Sometimes we need to store a **collection of values**
- Communication research almost always involves collections: many respondents, many headlines, many articles, many posts, or many model responses.
- **That's why lists and dictionaries are important for research**



The Dot . Usually Means: “Do Something With This”

- **headline** = the text we're working with
- . = use a tool that belongs to it
- **lower()** = make the text lowercase

```
#object and methods
"headlines".upper()
"I love communication research".replace("communication", "physics")
```

2. Python can help find information we need

Retrieve Information by Key

```
#Retrieve Information by Key
article["source"]
article["topic"]
article["sentiment"]
```

Indexing

- When we create a list of something, we have to know how to call up a specific item within the list.
- The process is called indexing
- Python starts counting at zero. This needs explicit attention because zero-based indexing is unintuitive for many beginners.
- What will headlines[1] return?

```
#Call for the order within the list
headlines[1]
```

Negative Indexing

- -1 = last item
- If there are 500 headlines, how would you retrieve the last one without knowing its index?

```
#Negative indexing
headlines[-1]
```

Sometimes We Want More Than One Item (“Slicing”)

- headlines[0:2] → items at indexes **0 and 1**
- headlines[:2] → exactly the same thing; omitting the start means “start from index 0”
- headlines[1:] → starts at index **1** and takes everything to the end

Strings Can Be Indexed Too

```
text = "journalism"

text[0]
text[:4]
len(text)
```

3. Python can repeat a task automatically

Loops: Tell Python to Repeat Something

- Imagine **500 headlines**. Instead of trying to clean headline, inspect comment, or changing the format one by one like the following :

```python
headline1.lower()
headline2.lower()
headline3.lower()
```

How A Loop Works

- Take one item
- Run the instruction
- Move to the next
- Stop when finished

```python
#the loop function for repeated instruction
for headline in headlines:
    print(headline)
```

```
AI changes journalism
Election coverage expands
New platform launches
```

Indentation Defines the Block

- Indented = repeated
- Not indented = after loop
- This is one reason blindly copying AI-generated code can FAIL: changing indentation changes the program's structure.

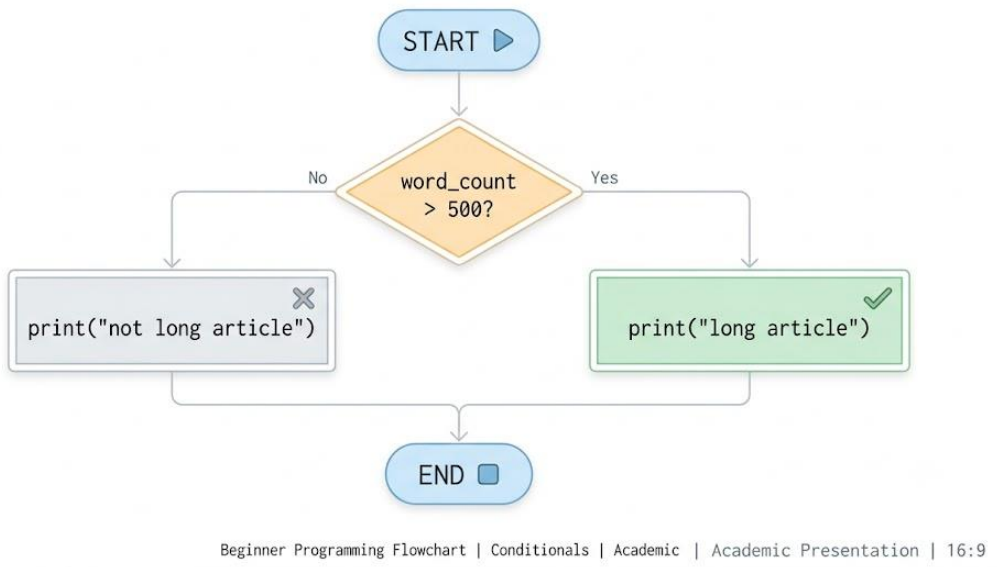
```python
#Indentation defines the block
for headline in headlines:
    clean = headline.lower()
    print(clean)

print("Finished")
```

4. Python Can Make a Simple Decision

if = Tell Python: “If This Is True, Do This”

```
#Conditioning
word_count = [500, 1000, 1500]
for word_count in word_count:
    if word_count > 500:
        print("long article")
    else:
        print("short article")
```

```
short article
long article
long article
```

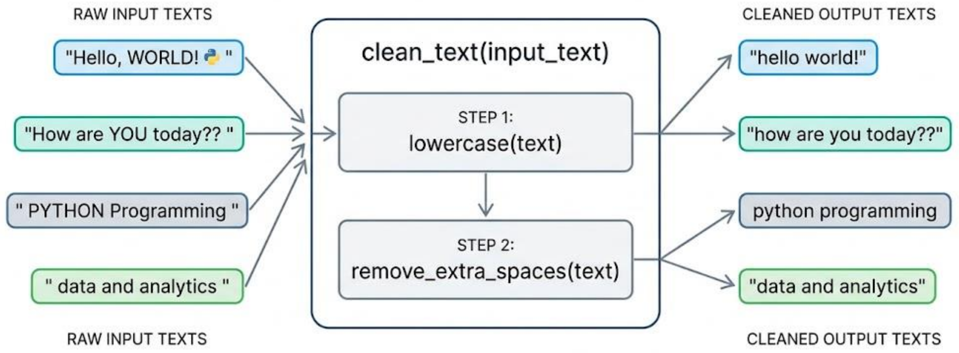


5. Python Can Save Instructions to Use Again

Functions: Save Instructions We Want to Use Again

Python

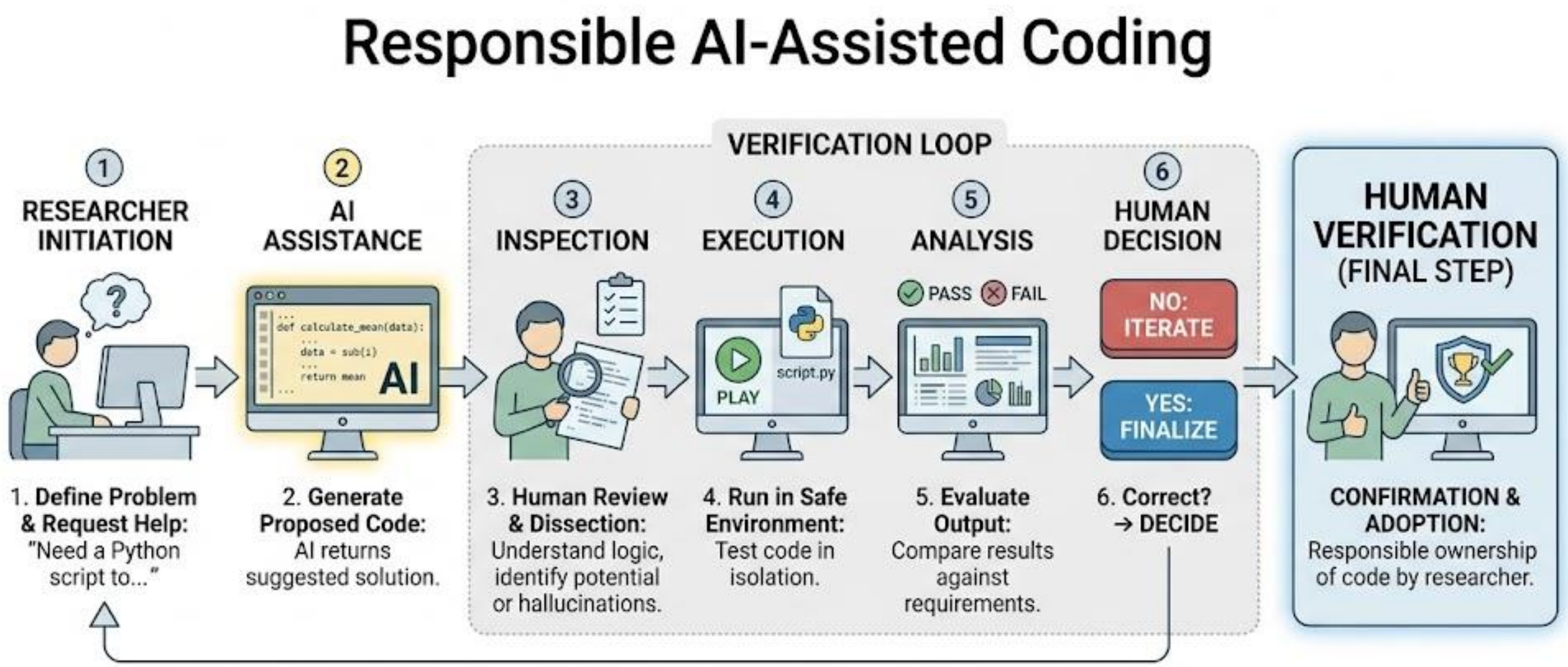
```python
# Save a set of instructions as a function
def clean_text(text):
    text = text.lower()
    return text.strip()


# Some example text
texts = [
    " Hello, WORLD! ",
    " PYTHON Programming ",
    " SPORTS Journalism ",
    " Data and Analytics "
]


# Use the same function on every item
for item in texts:
    cleaned_text = clean_text(item)
    print(cleaned_text)
```



Why Functions Matter for Research

- Consistent procedure
- Less duplicated code
- Easier revision
- Easier testing
- Better reproducibility

AI-generated code is a proposed solution, NOT a verified solution.



Assignment 2: Due in Class During Next Thursday's Lab

1. Choose a Research Question

SPORTS JOURNALISM / SPORTS MEDIA

- How does social media change the way sports journalists report breaking news?
- How does social media coverage influence fans' perceptions of athletes?
- How does sports media coverage differ between men's and women's sports?

SOCIAL MEDIA & MEDIA EFFECTS

- How does short-form video affect how people consume news?
- How does social media use affect trust in news and information?
- How does social media content influence people's perceptions of public figures?

1. Choose a Research Question

STRATEGIC COMMUNICATION / MARKETING

- How does influencer marketing affect consumers' trust in brands?
- How does a company's response to a social media crisis affect public opinion?
- How does AI-generated advertising affect perceptions of brand authenticity?

MEDIA, SOCIETY & PROFESSIONAL ISSUES

- How does news coverage influence public perceptions of high-profile court cases?
- How does media literacy affect people's ability to identify misinformation?
- How does generative AI affect students' research and writing practices?

2. Prepare Your Abstract Set

- Groups of 4-5
- Choose one research question
- Find **3 academic abstracts** PER PERSON
- Resources: Google Scholar. Select Only Peer-Reviewed Articles
- Save as one .csv file
- Bring the file to the Week 4 lab

What Are Peer-Reviewed Articles?

- Research articles evaluated by other experts before publication
- Usually published in academic or scholarly journals
- Look for articles that include an **abstract, methods, results, and references**
- Find them through **Google Scholar** or the **USC Libraries databases**
- For this activity, use **peer-reviewed journal articles only**
- Avoid news articles, blogs, websites, reports, theses, and book chapters

3. Save Your Abstracts Like This (One File Per Group)

Your CSV should contain:

Group member	abstract
	This study examines how journalists...
	Using survey data, this study explores...
	This paper investigates physicians...

4. Your Abstract Set Should Be Mixed

Try to include:

- **One clearly relevant:** Directly addresses your question
- **One possibly relevant:** Related construct, population, or technology
- **One probably unrelated:** Similar keywords, different research problem

Example question: **How does social media coverage influence fans' perceptions of athletes?**

- **Clearly relevant:**
  Social media coverage of athletes + fan perceptions
- **Possibly relevant:**
  Athletes' own social media use + fan engagement
- **Probably unrelated:**
  Social media advertising + sporting event attendance

Next Class: ZOOM LAB

- Zoom link for this Thursday is posted on Blackboard.
- Do not come to class next Thursday